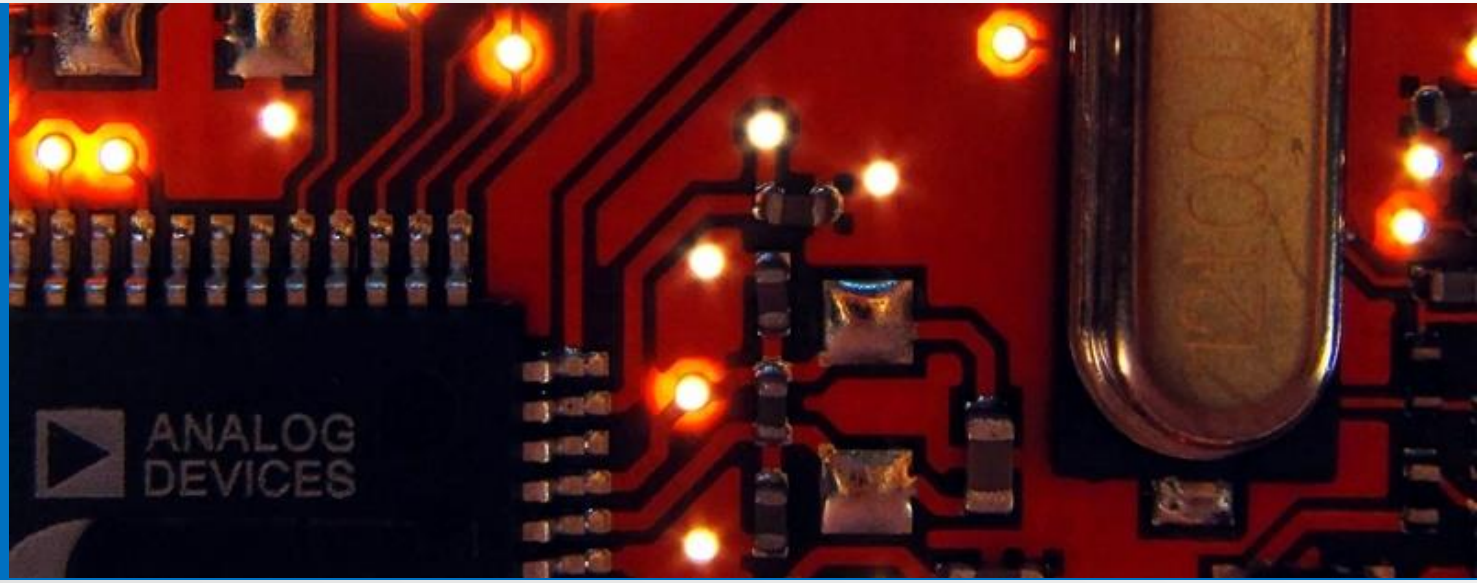
DEEA / CIL



Curs Introductiv

V 2022

Disclaimer

- *Slide-urile acestea reprezintă un rezumat.*
- *Slide-urile nu acoperă toate explicațiile, simulările, demonstrațiile oferite în timpul cursului.*
- *Slide-urile nu limitează în vreun fel materia cerută la examen.*
- *Pentru varianta completă, sunteți așteptați la curs*

Miercuri 08:00 - 10:00 PR001

Joi 08:00 – 10:00 AN030

(pe par) Marti 18:00 – EC105

Cuprins

1. Intro
2. Structură punctaj
3. Introducere / despre sisteme
4. Despre curs / cuprins
5. Reguli
6. Next time

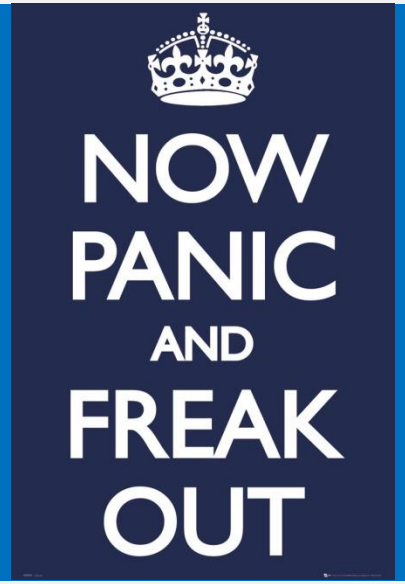
Resurse EA

- curs (online), laborator (online)
- acs.curs.pub.ro
  - slide-uri curs, îndrumare laborator
  - forumuri, feedback, etc
- CISL41 Hardware Design & Robotics Laboratory

Așteptări de la studenți

- Să fiți prezenți & atenți ☺
- Să fiți implicați, să puneți întrebări, să discutați
- Să nu lăsați (prea multe) neclarități în urma unui curs
- Folosiți-vă de “resursa” profesor

Test



Test

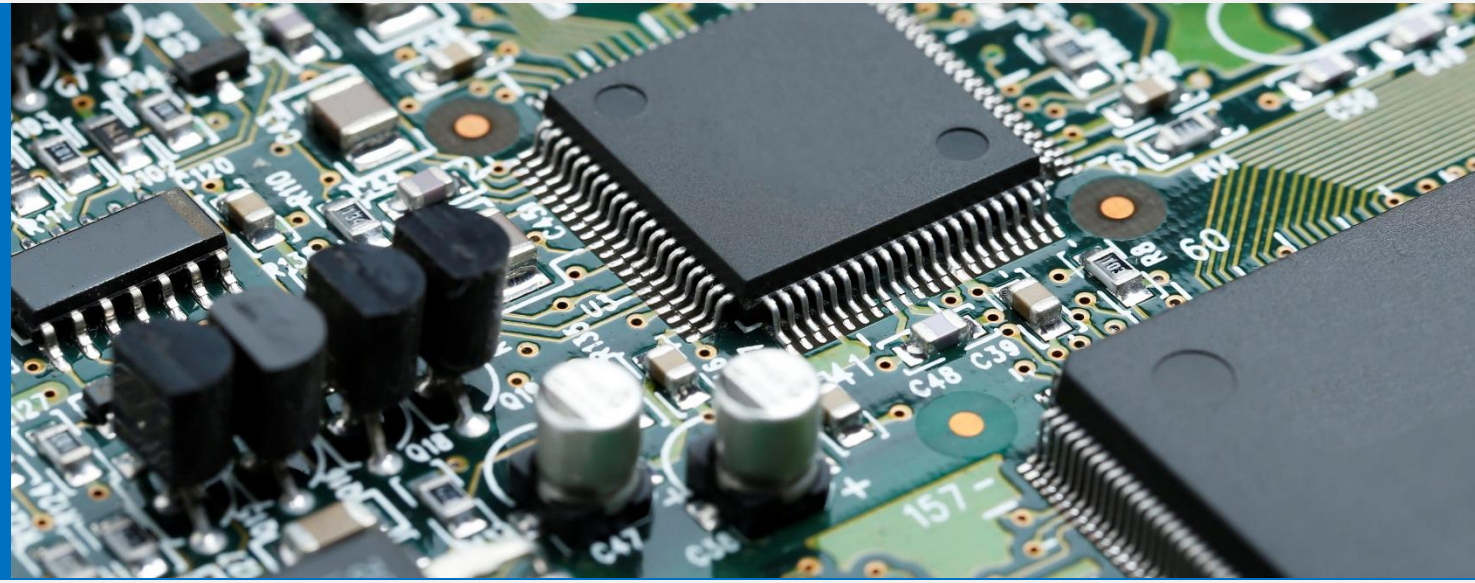
menti.com

2875 7532

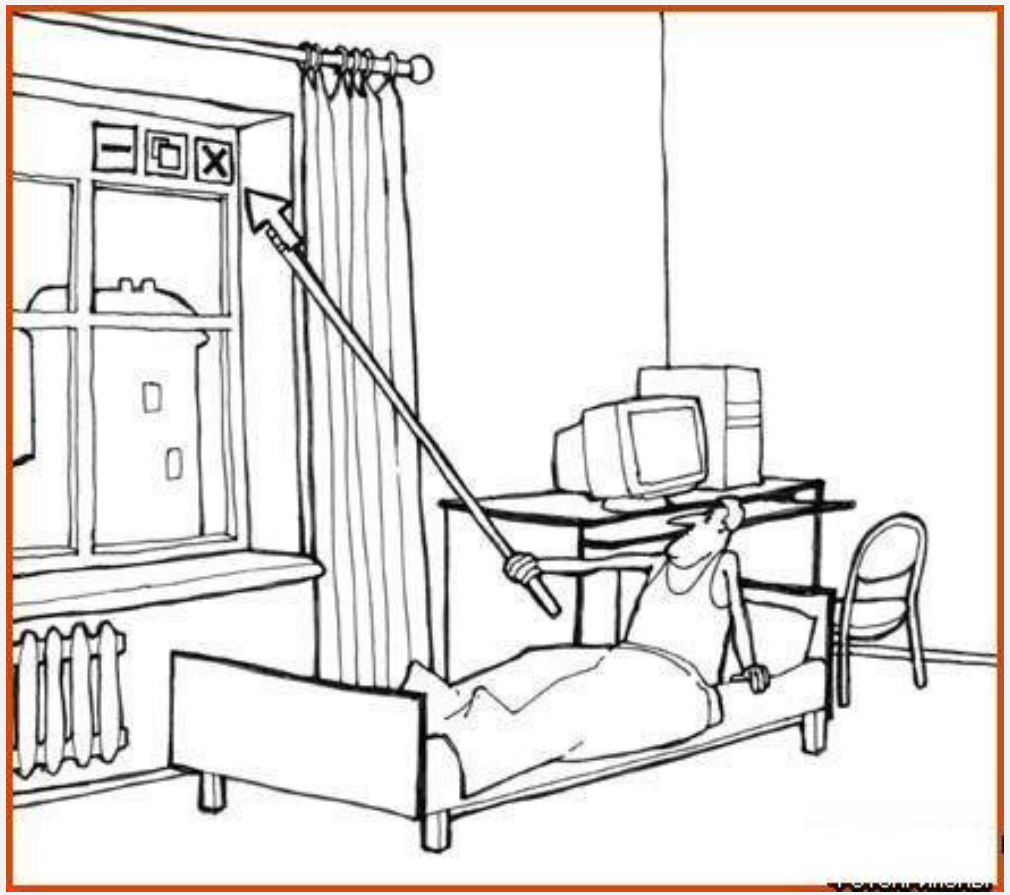
https://www.menti.com/alk8e2jd5r38



Despre



Inginer

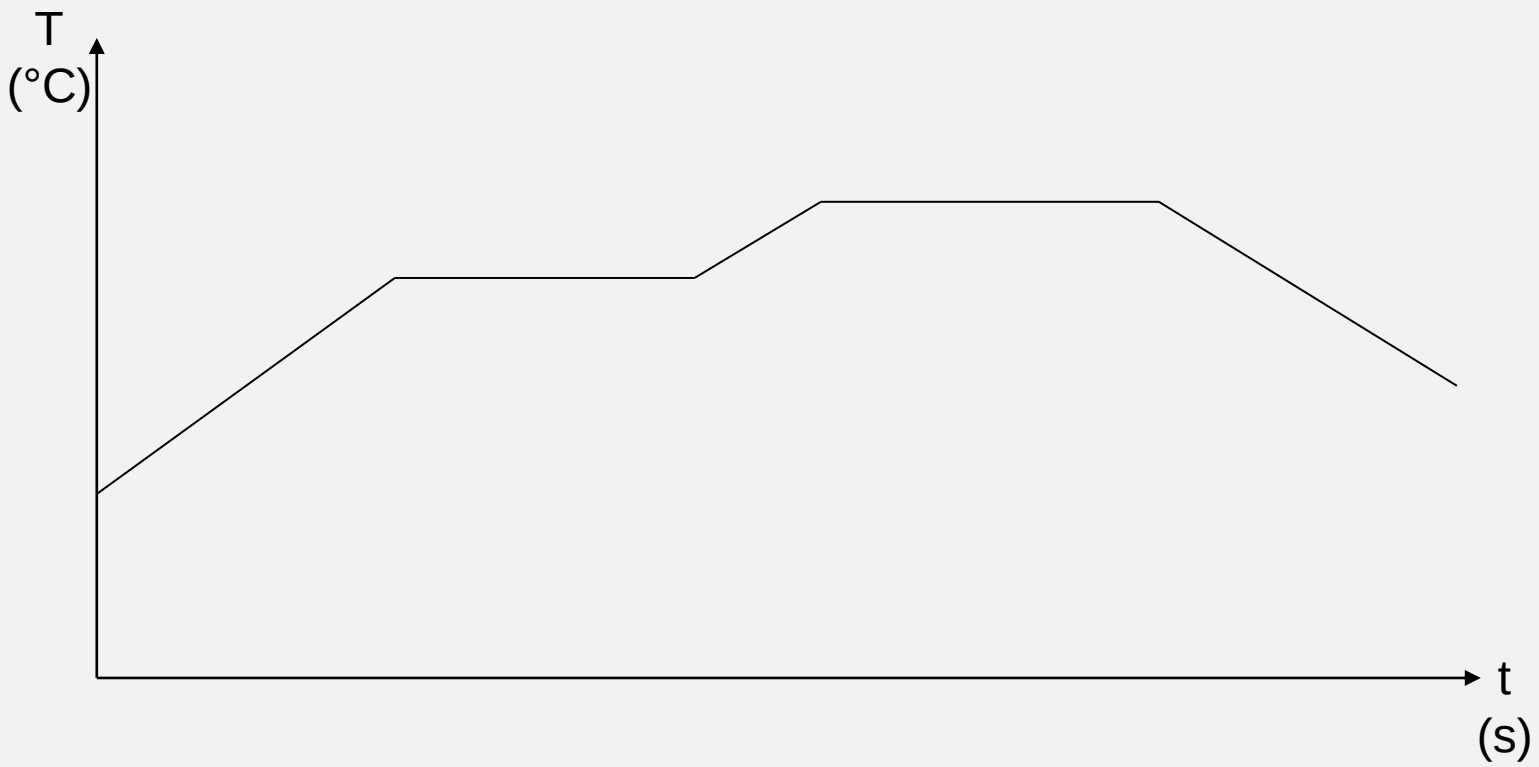


scientific understanding of the natural world

>> used to invent, design, and **build things**
>> to **solve problems** and achieve **practical goals**

© https://study.com/academy/lesson/what-is-engineering-definition-types.html

Inginer



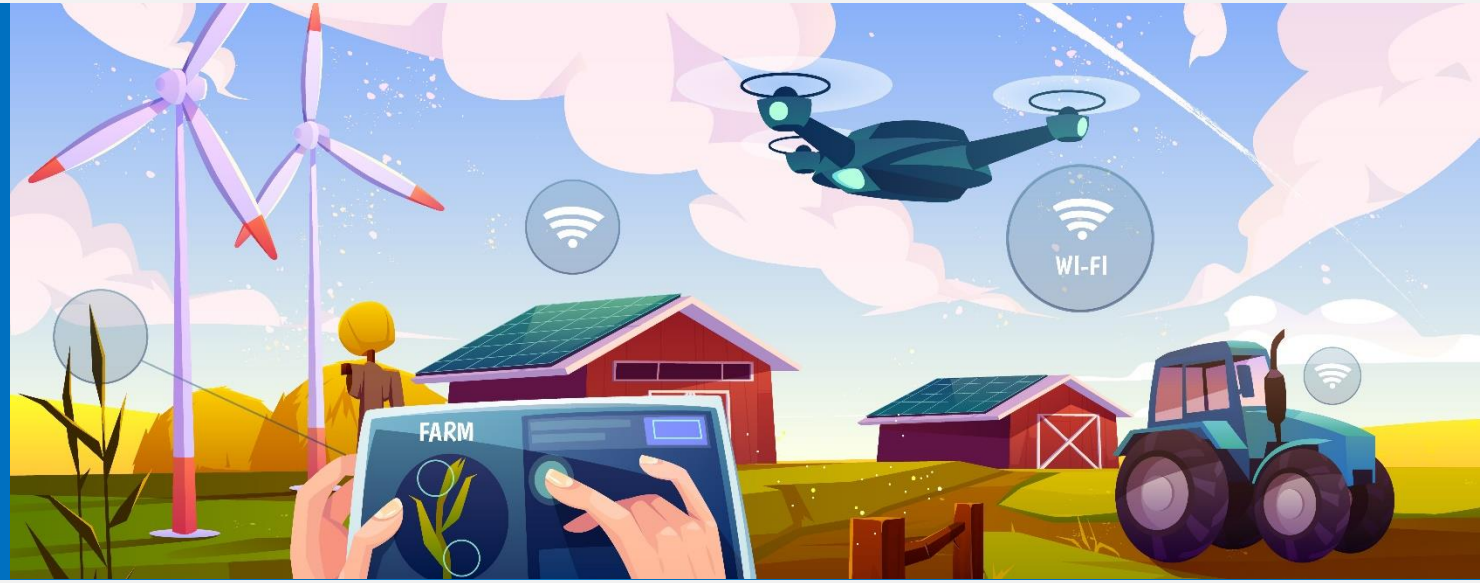
Electronica @ CTI

Proiectarea VLSI
Sisteme încorporate
Sisteme de achiziții de date
Proiectarea cu microprocesoare
Structura și organizarea calculatoarelor
Electronică digitală
Dispozitive electronice și electronică analogică
Fizică
Electrotehnică

Electronica @ AIA

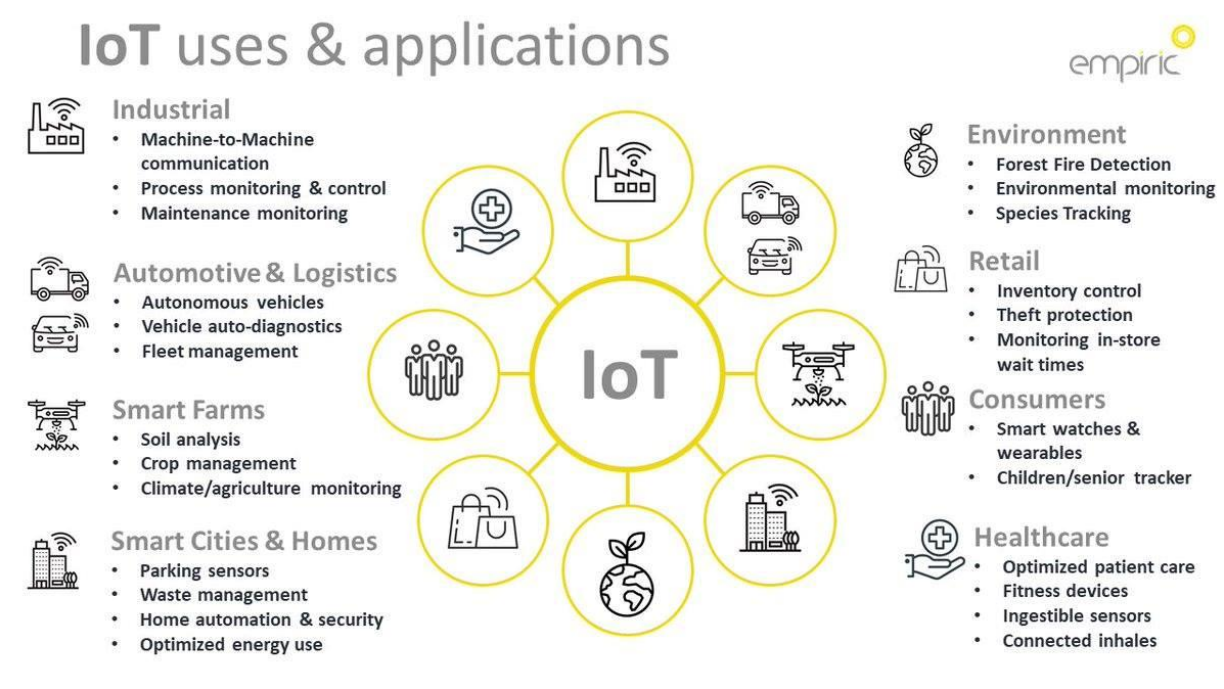
Transmisia datelor
Sisteme cu microprocesoare
Arhitectura calculatoarelor
Măsurări şi traductoare
Instrumentatie
Maşini electrice şi acţionări
Electronică digitală
Circuite electronice liniare
Fizică
Electrotehnică

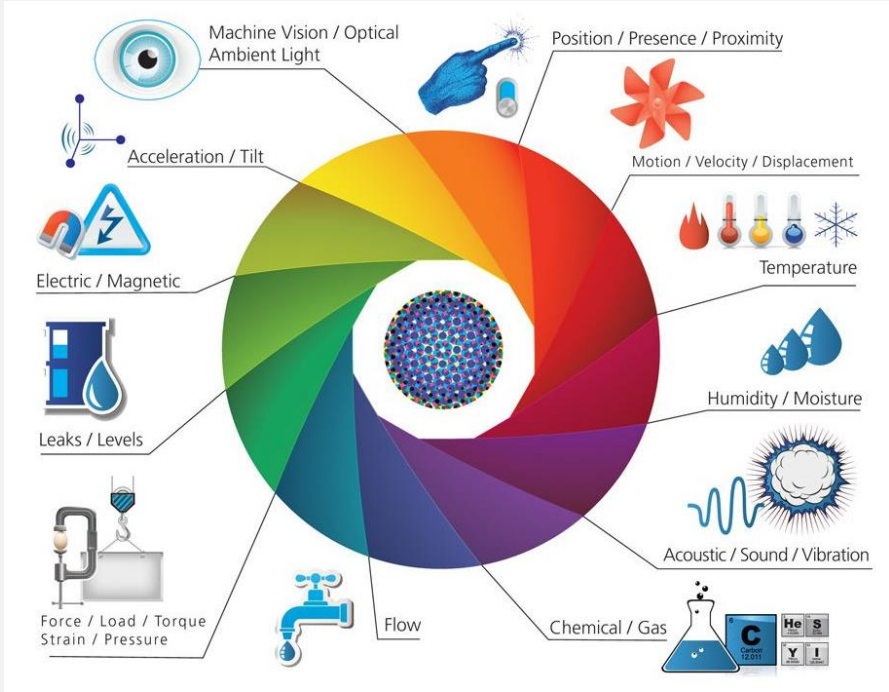
IoT



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IoT

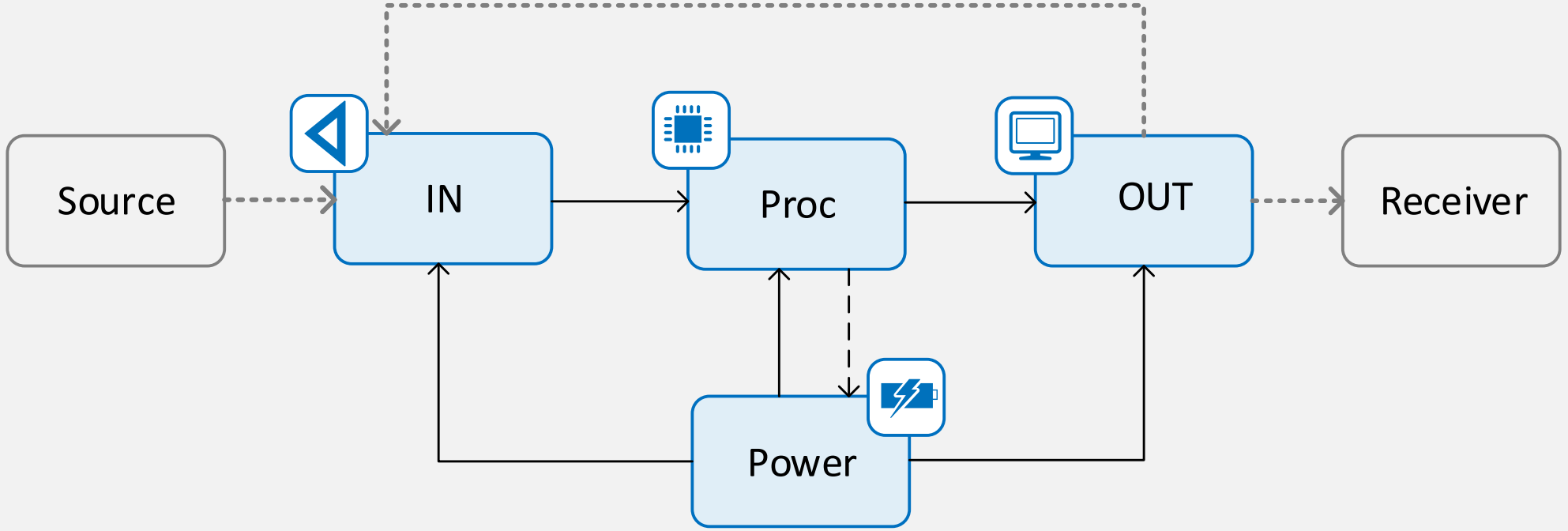




© https://www.facebook.com/empiricsolutions/photos/the-global-iot-application-platform-and-services-revenue-is-forecast-to-amount-t/2947857835263186/

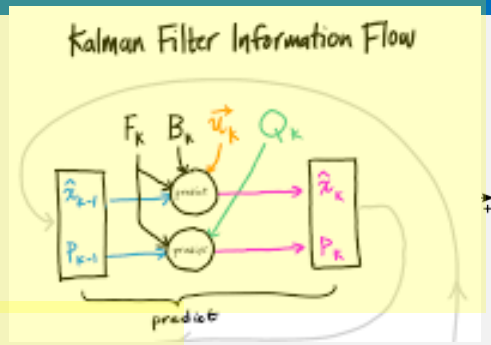
© https://www.postscapes.com/what-exactly-is-the-internet-of-things-infographic/

De ce Electronică?

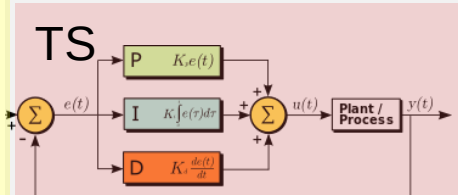


(drone ex.)

SII



TS

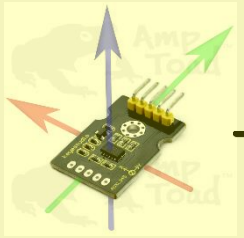


ELTH

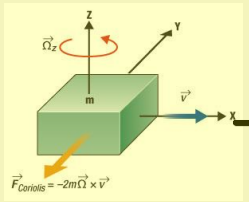


EEA

Custom sensors



Accelerometer



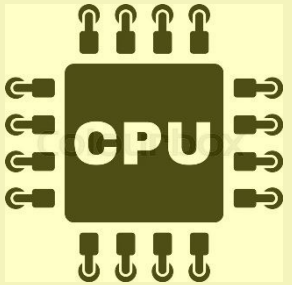
Gyroscope

AD





ED

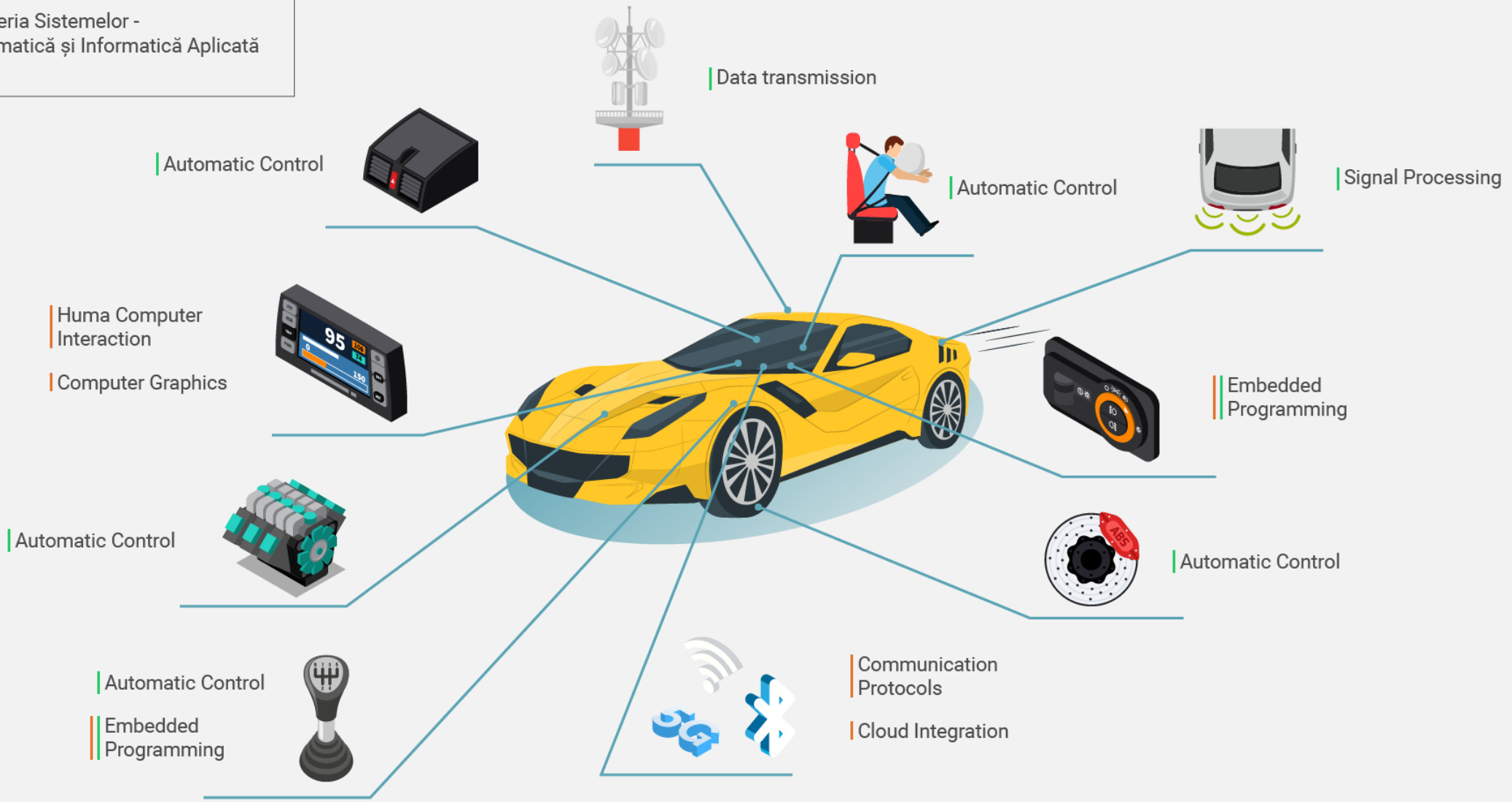


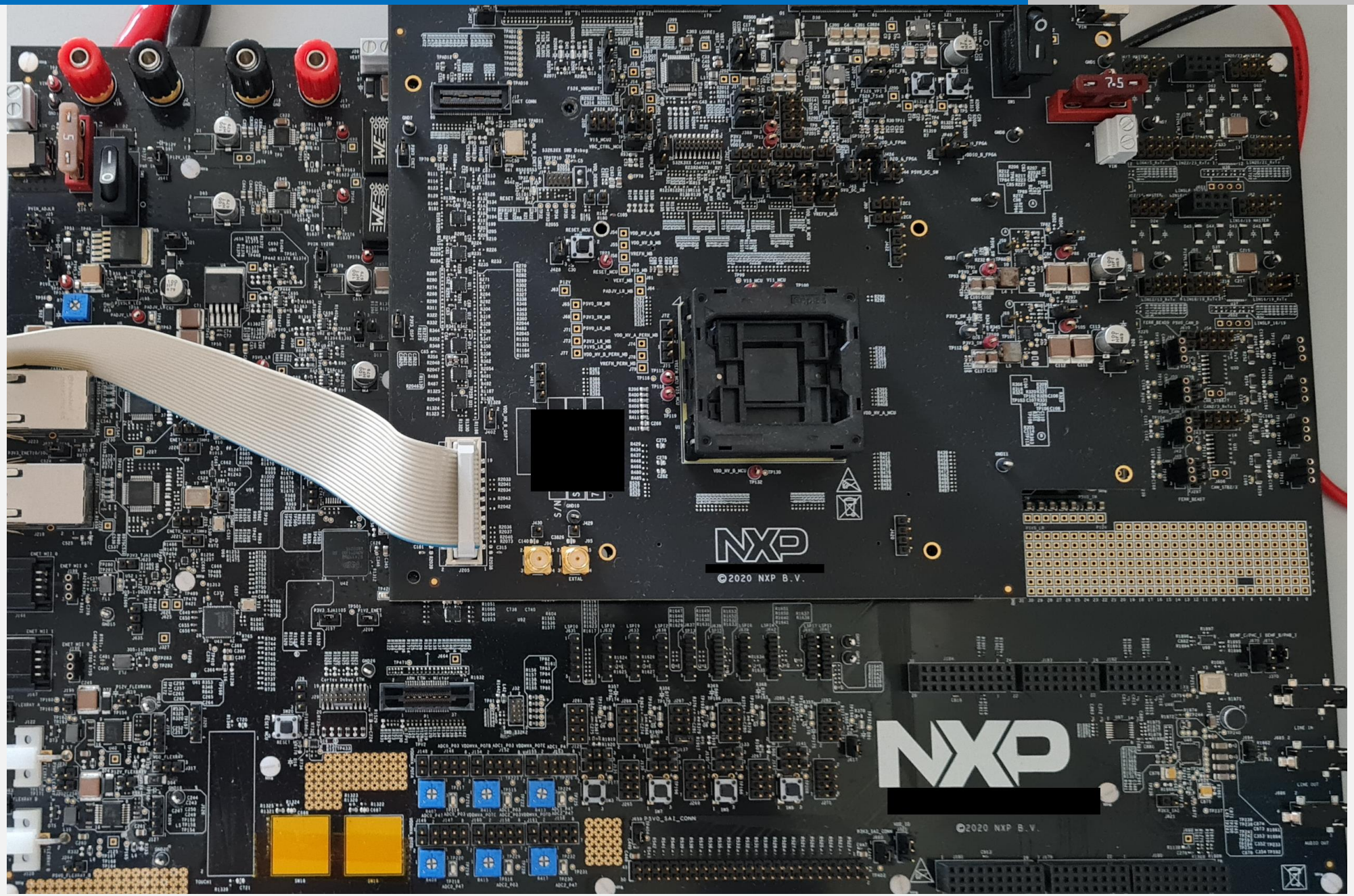
EEA



Calculatoare și
Tehnologia Informației

Ingineria Sistemelor -
Automatică și Informatică Aplicată



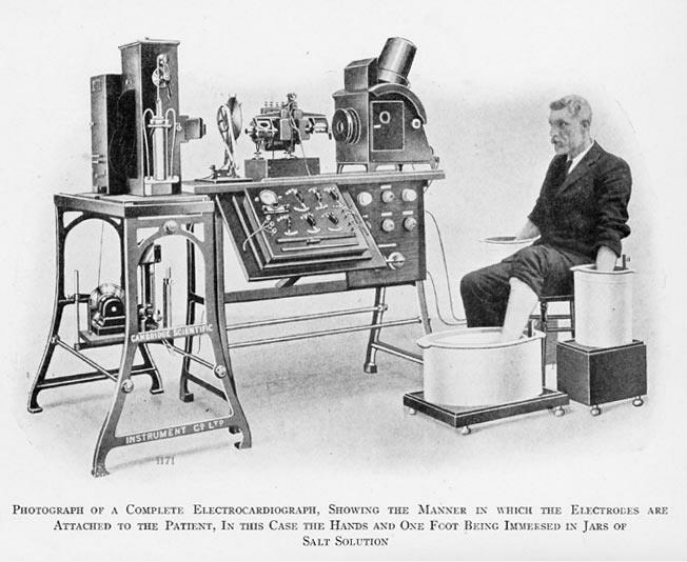

NXP
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NXP
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Unde suntem

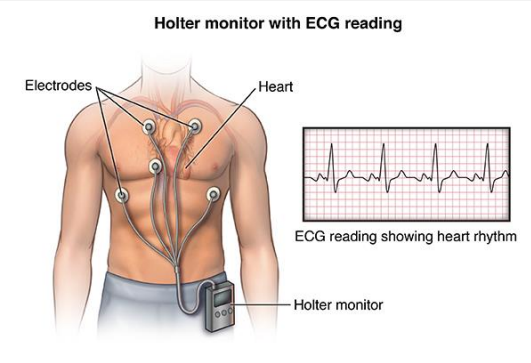




Unde suntem



PHOTOGRAPH OF A COMPLETE ELECTROCARDIOGRAPH, SHOWING THE MANNER IN WHICH THE ELECTRODES ARE ATTACHED TO THE PATIENT, IN THIS CASE THE HANDS AND ONE FOOT BEING IMMERSED IN JARS OF SALT SOLUTION

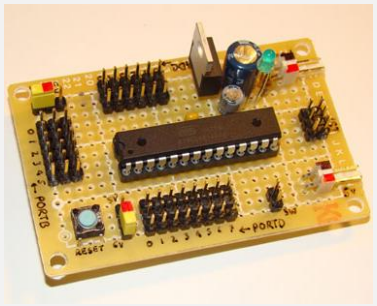




© https://en.wikipedia.org/wiki/File:Willem_Einthoven_ECG.jpg

https://www.samsung.com/us/apps/samsung-health-monitor/

Opțiuni, peste tot opțiuni

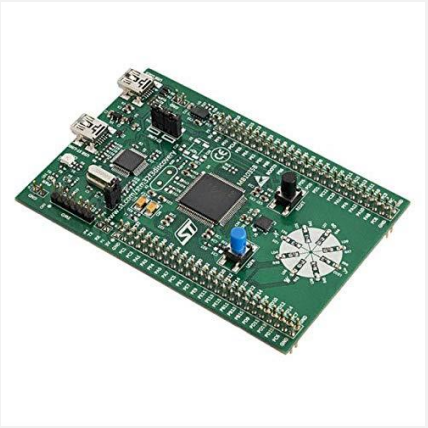




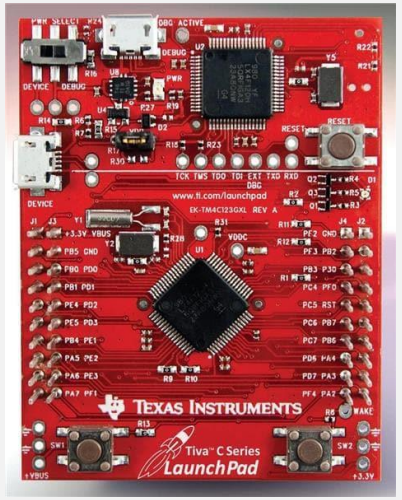
ChipKIT
$30
PIC



Arduino
Nano Every, Nano 33IOT, Nano 33 BLE, Nano 33 BLE Sense
$9.90 - $29.50

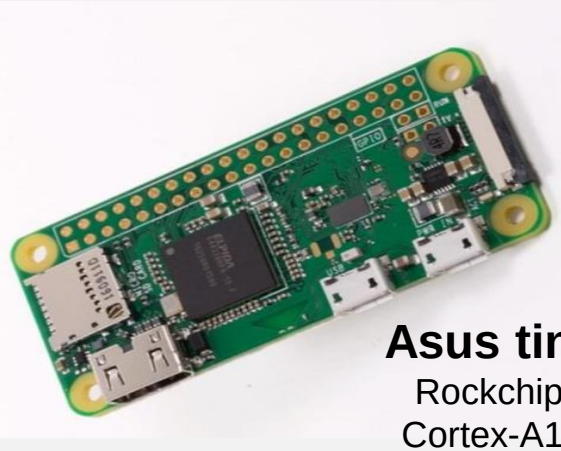


STM32 F3 Discovery
$20
ARM Cortex M4



LaunchPad
$15
ARM Cortex M4F





Asus tinker board
Rockchip RK3288 (4x Cortex-A17 @ 1.8GHz)

Mali-T760

2GB LPDDR3 RAM

89$

Echipa de Electronica



Asistenții



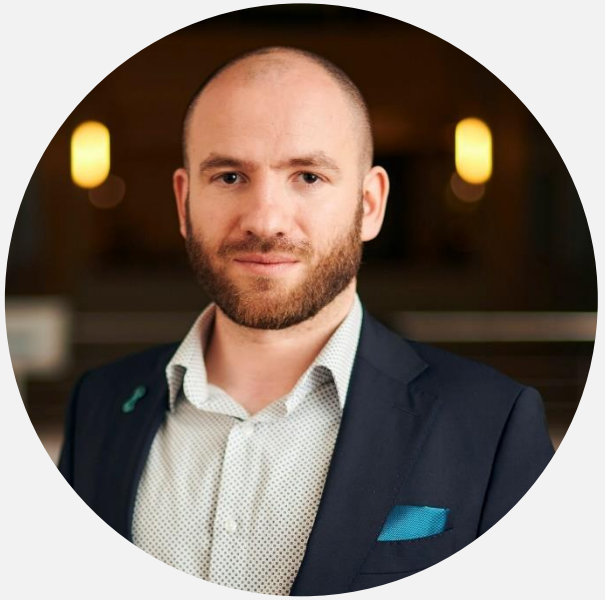


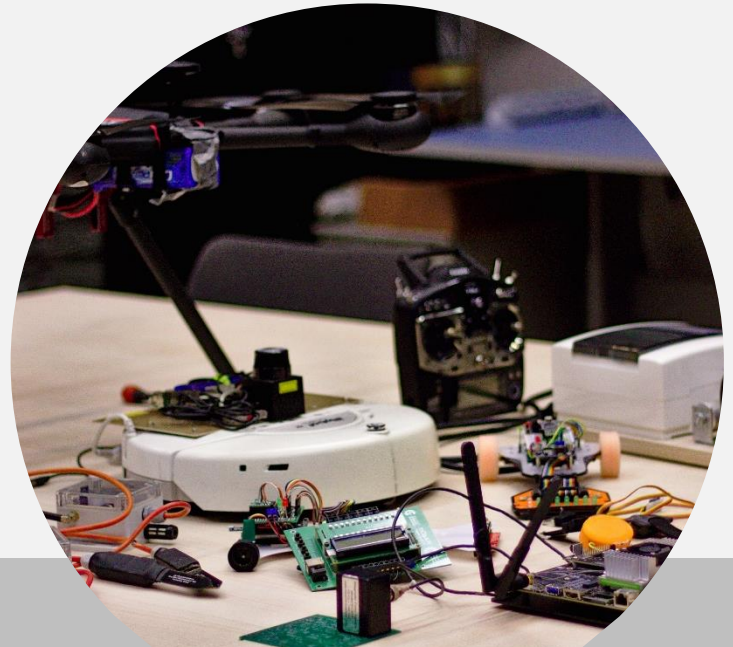


About me (Daniel Rosner)

- Cursuri
  - (II) Electronică Analogică
  - (III) Proiectarea cu Microprocesoare
  - (MI/MII) Sisteme Fizico-Cibernetice
- VZ & PoliFest
- Innovation Labs
- OxidOS Automotive
- Concursuuuuri (tech)
- Tech
  - IoT, Automotive, Energy, MedTech
- C310









Structură punctaj



Structură punctaj - AB

- 3p – teste in timpul semestrului
- 2p – laborator
- 1p – tema
- 4p – examen final

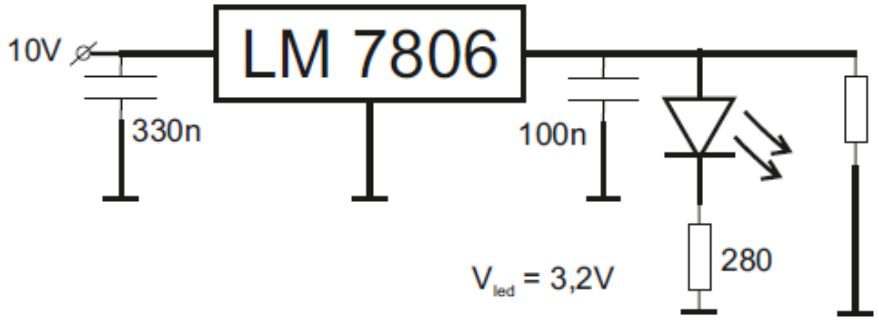
Structură punctaj - CC

- 2p – teste in timpul semestrului
- 2p – laborator
- 2p – proiect
- 4p – examen final

Structură examen final

- (25-30p) 20 – 25 subiecte cu raspuns scurt x 1-2 puncte
- (10-15p) 1 subiect scris aplicat, cu mai multe subpuncte -
- *(veti avea voie cu 1 foaie A4 față-verso, scrisă de mână, cu orice considerați necesar pentru examen)*

(exemple subiecte)



S03

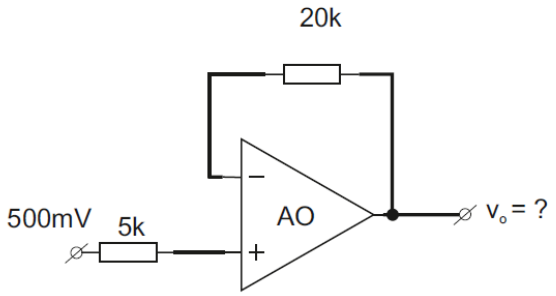
a) I_{Sursa} = ?

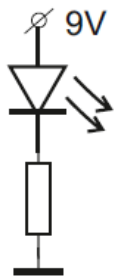
b) Care este valoarea puterii totale consumate de la sursa de 10V?

Se conectează o sursă de semnal cu Zo = 1k și Uo = 100mV la intrarea unui amplificator de tensiune.

Tensiunea măsurată la intrarea amplificatorului este 99mV.

Să se calculeze, cu o eroare de 10%, impedanța de intrare.





V_{led} = 2,8V

Să se propună o valoare pentru R, astfel încât I_{Led} să fie cuprins între 10 și 20 mA.

Reguli



Reguli pentru cursul de EA

Reguli promovare

- Condiții minime pentru promovare:
  - 50% parcurs
  - 50% examen
  - minim 4 (din 6) prezente @ laborator
    - punctaj laborator – va fi prezentat în cadrul laboratorului 1.

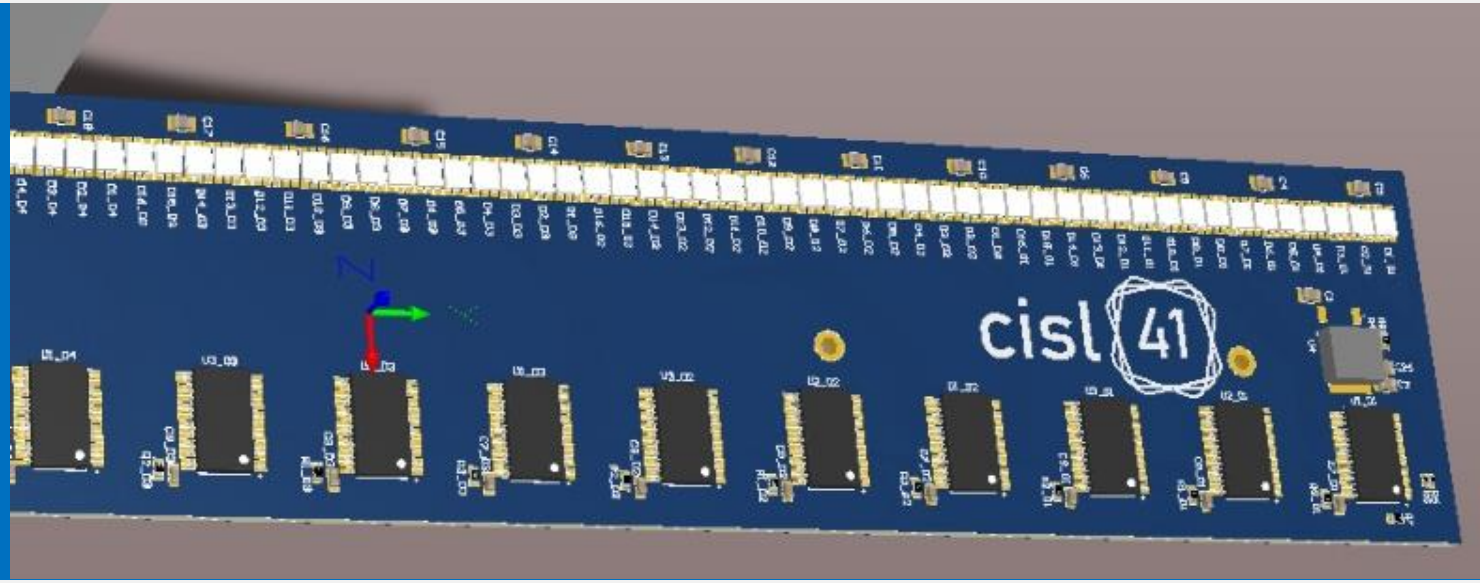
Extra

- **Bonus** pentru rezultate la concursuri & activități
  - până la 1,5 puncte pentru concursuri de profil hardware
  - până la 1 punct pentru orice tip de concurs tehnic
  - până la 0,5 puncte bonus pentru implicare în activități studențesti de voluntariat

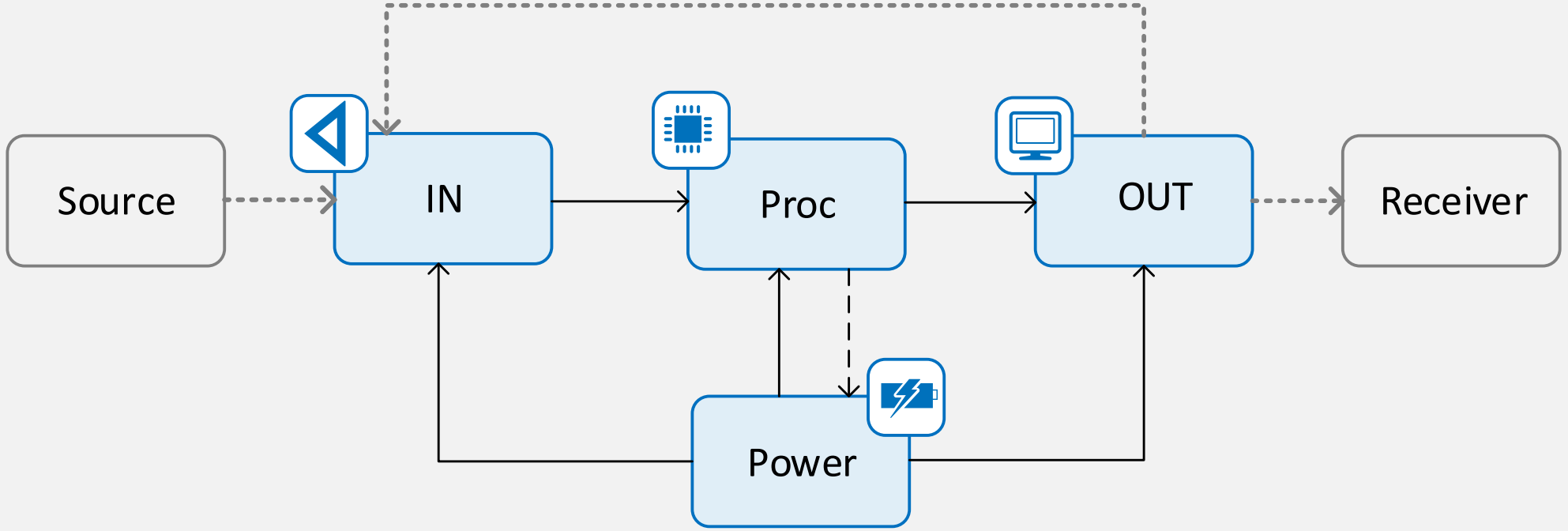
➢ *mail în presesiune cu Subject:*

➢ ***[Bonus_EA] Nume_Prenume_32xAB / 32xCA***

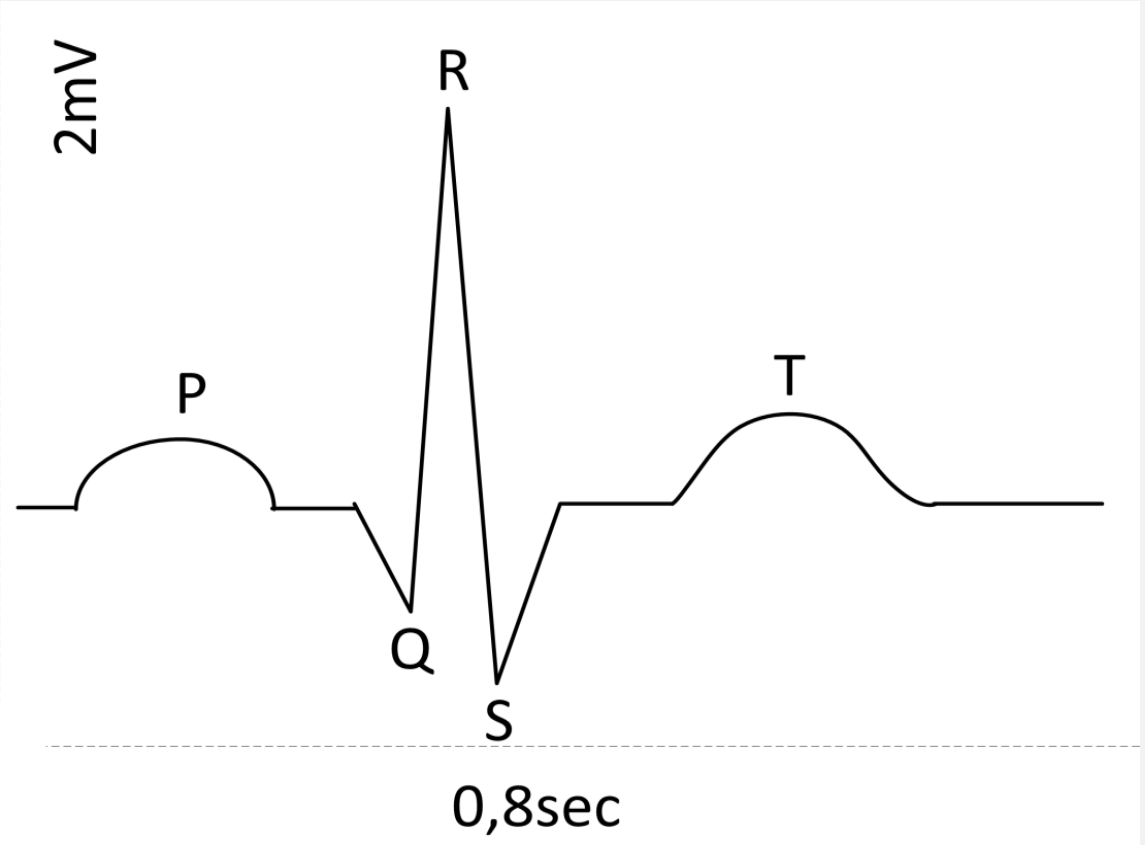
Electronica Analogica



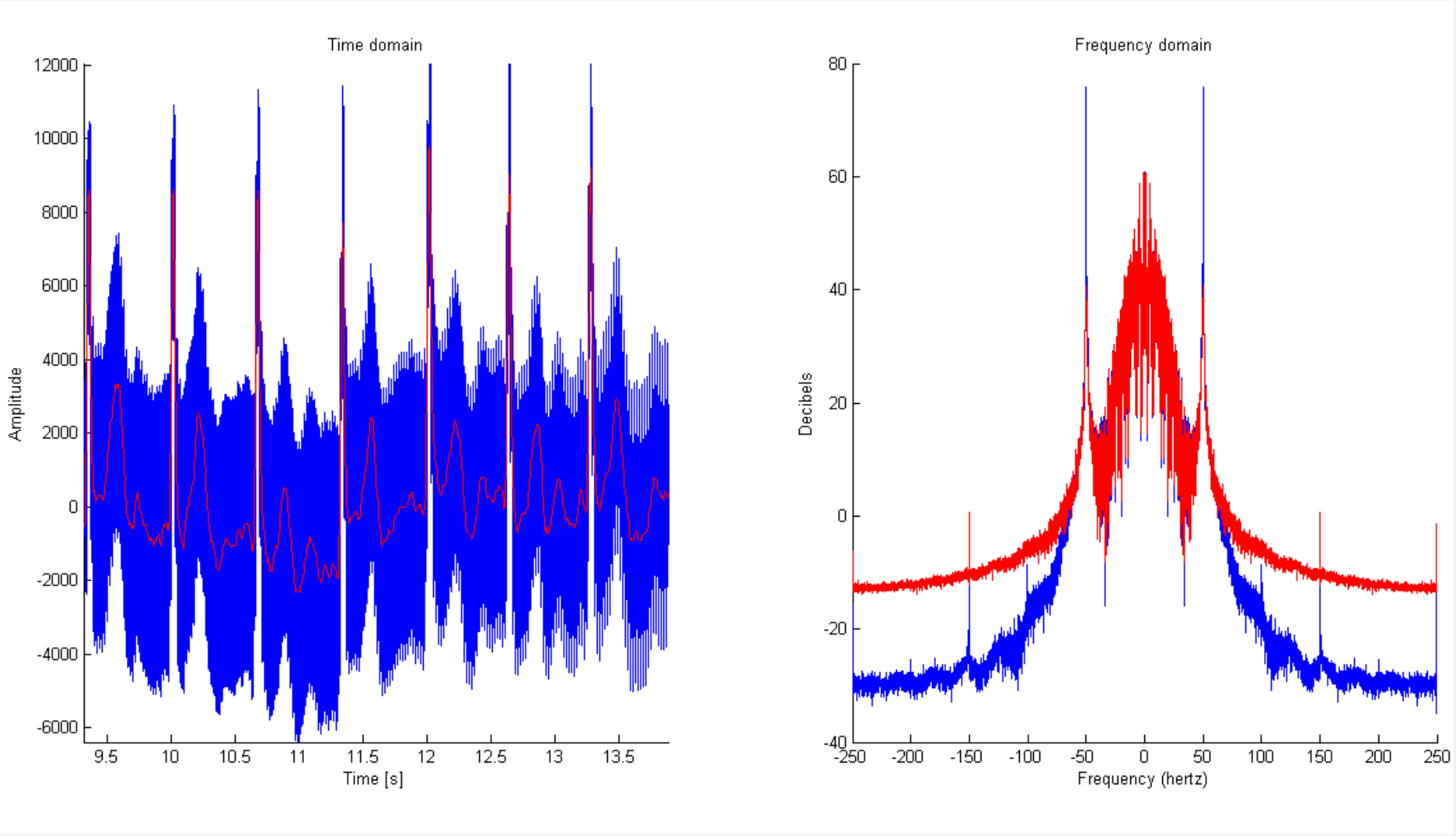
De ce Electronică?



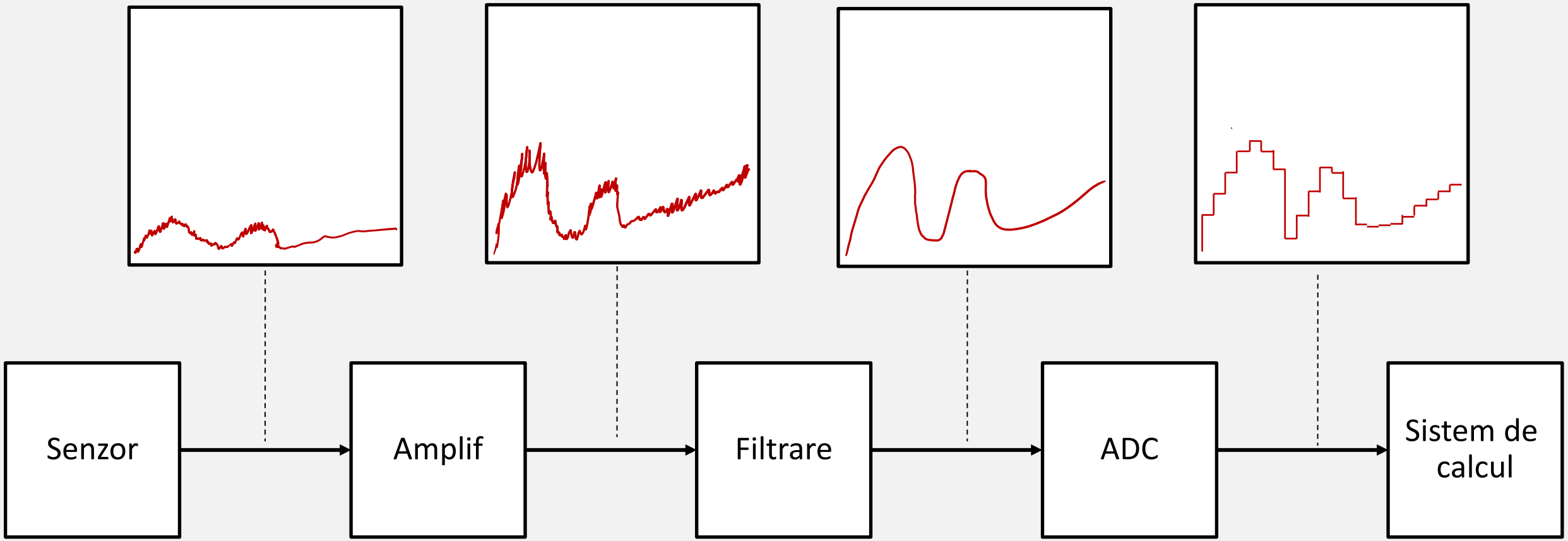
De ce Electronică?



De ce Electronică?



Schema generala sistem de AD

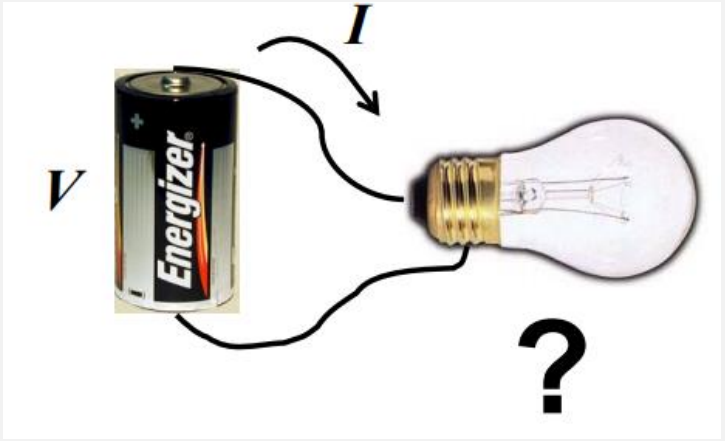


De ce Electronică?

- Tensiunea la borne?
- Ce se întâmplă când scade sub un prag recomandat?
- Supracurent?
- Cum se reîncarcă?
- Capacitate ramasă?



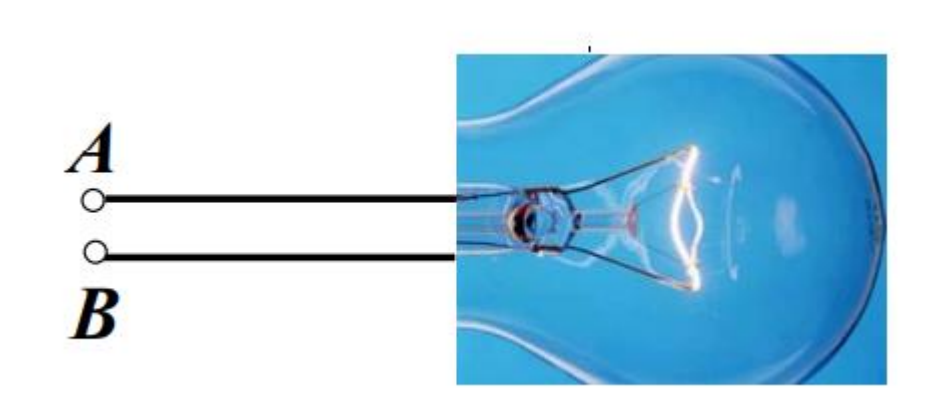
Abstractizare



$$\nabla \times E = -\frac{\partial B}{\partial t} \qquad \oint E \cdot dl = -\frac{\partial \phi_B}{\partial t}$$

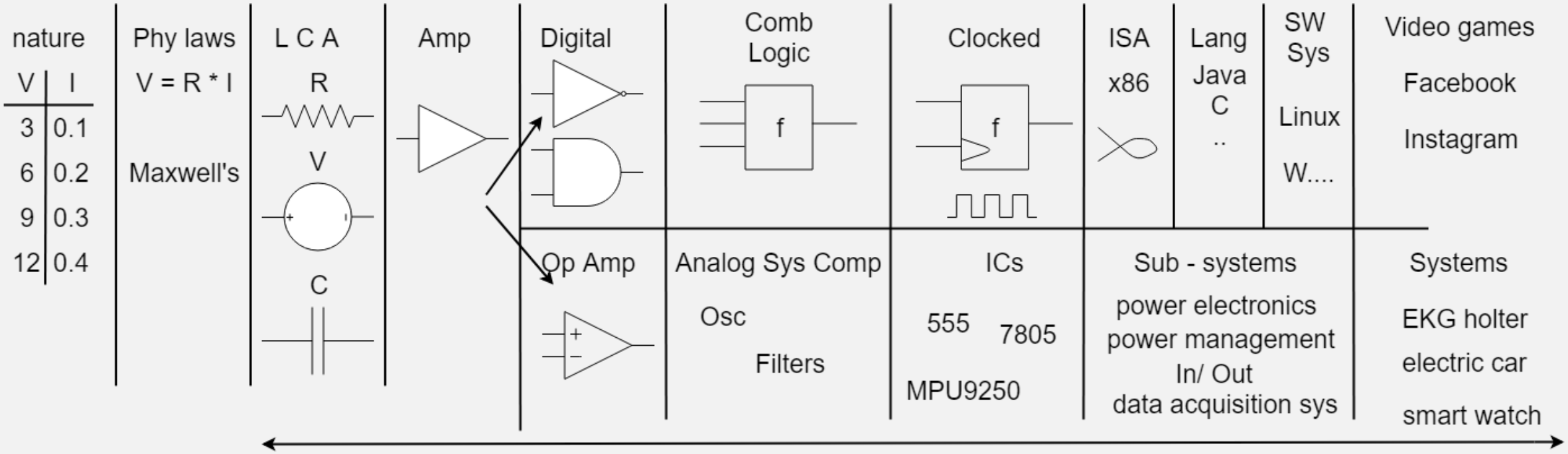
© Anant Agarwal and Jeffrey Lang, course materials for 6.002 Circuits and Electronics, Spring 2007. MIT

Abstractizare



© Anant Agarwal and Jeffrey Lang, course materials for 6.002 Circuits and Electronics, Spring 2007. MIT

Abstractizare



© Anant Agarwal and Jeffrey Lang, course materials for 6.002 Circuits and Electronics, Spring 2007. MIT

Pe scurt (EA & ED)

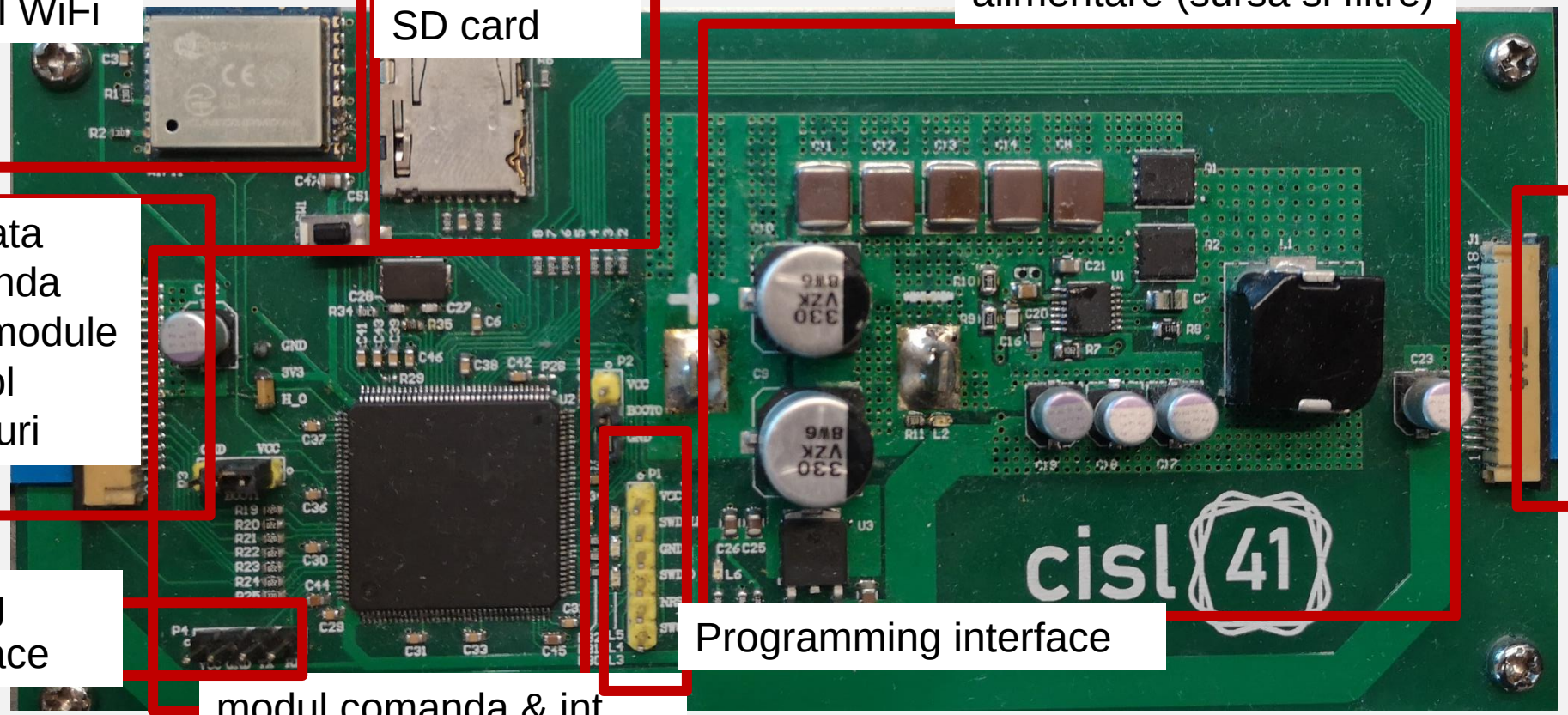
- **Proiectare/ modelare/ testare/ analiză** - dispozitive ce realizează controlul curentului electric
- Diferite scopuri:
  - acționări elemente de execuție / afișare
  - “citire” semnale / diferite mărimi fizice
  - prelucrare semnale
  - adaptare semnale
  - adaptare alimentare

Electronică Analogică

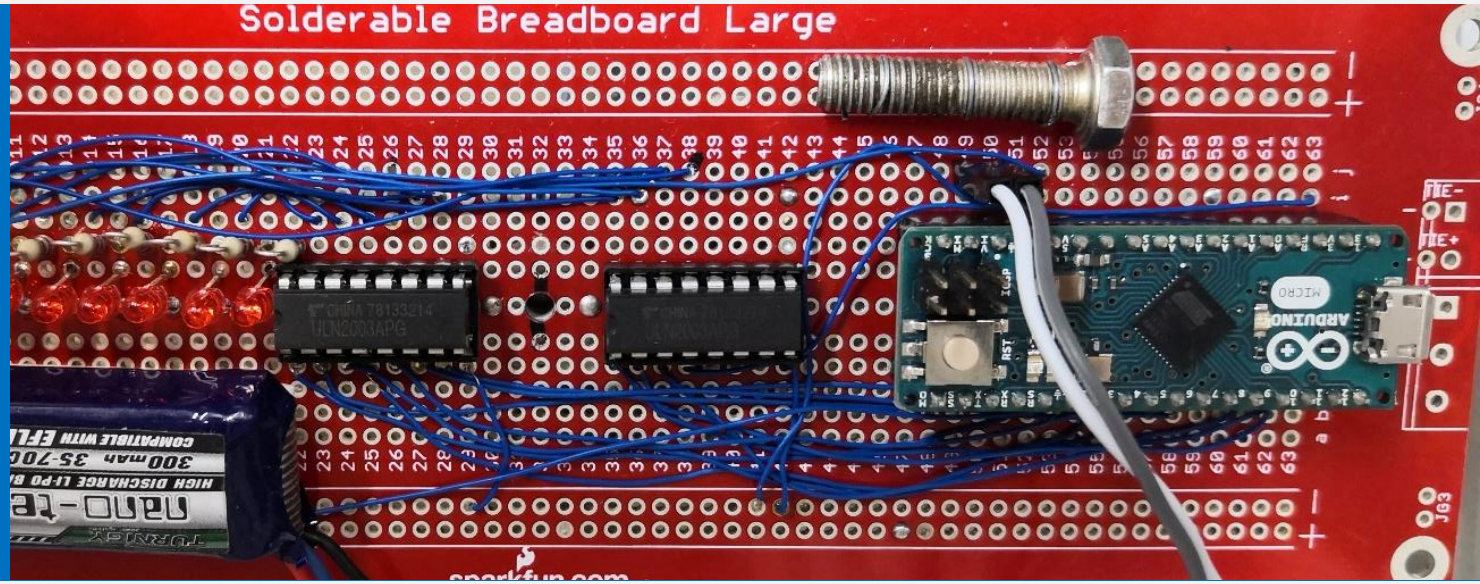


Câteva definiții

- Circuit electronic – modul / sub-modul / ansamblu de componente electronica și interconexiuni îndeplinind o funcție electrică

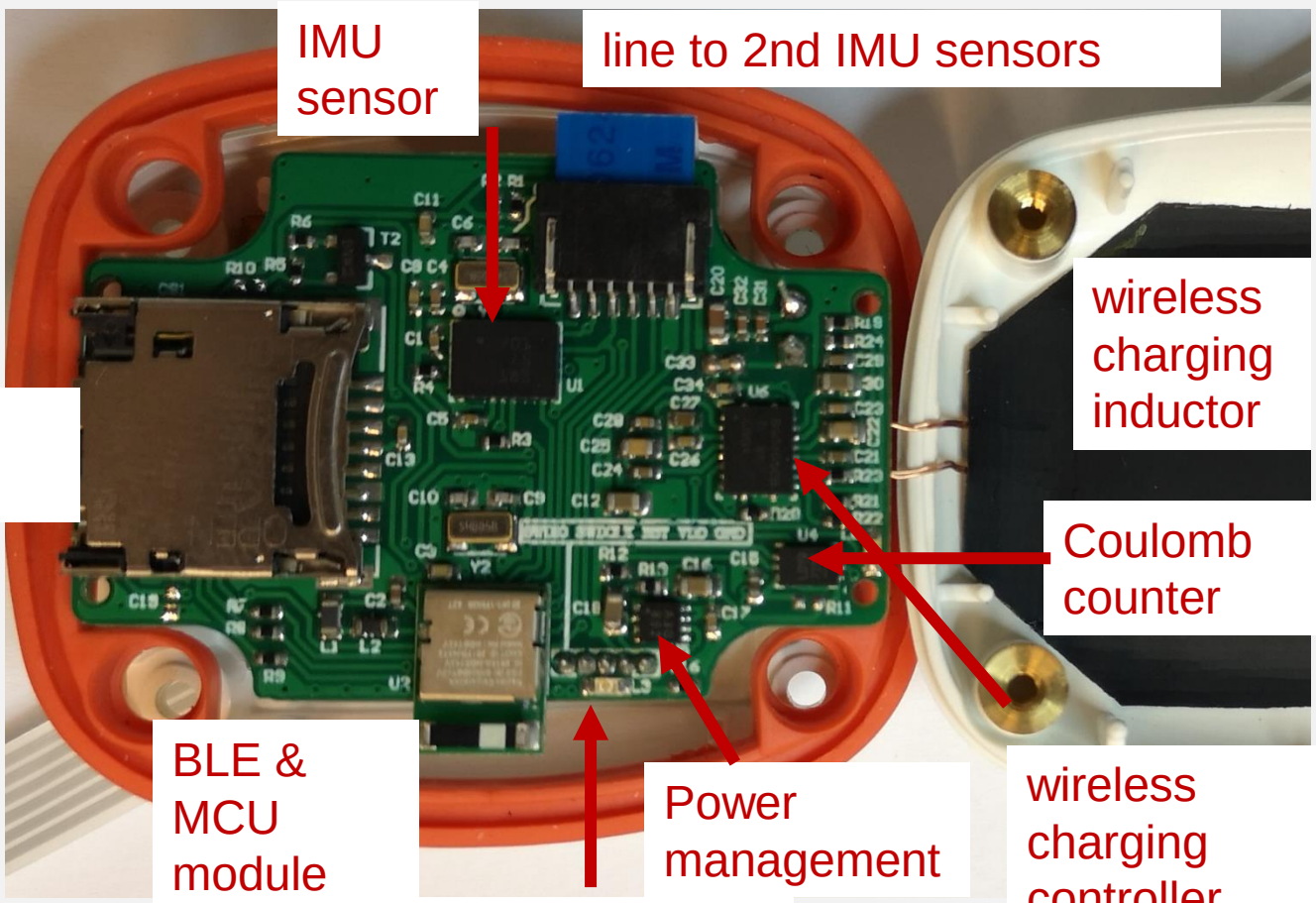


Discutie

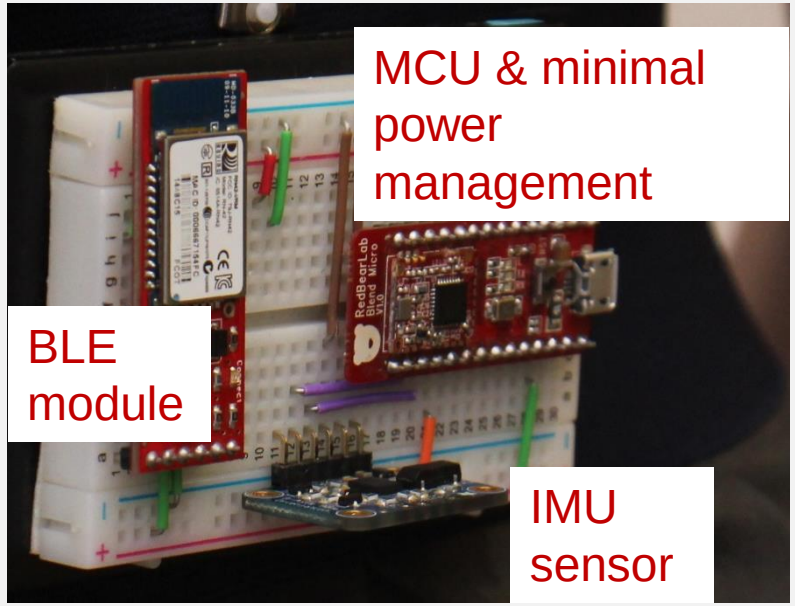


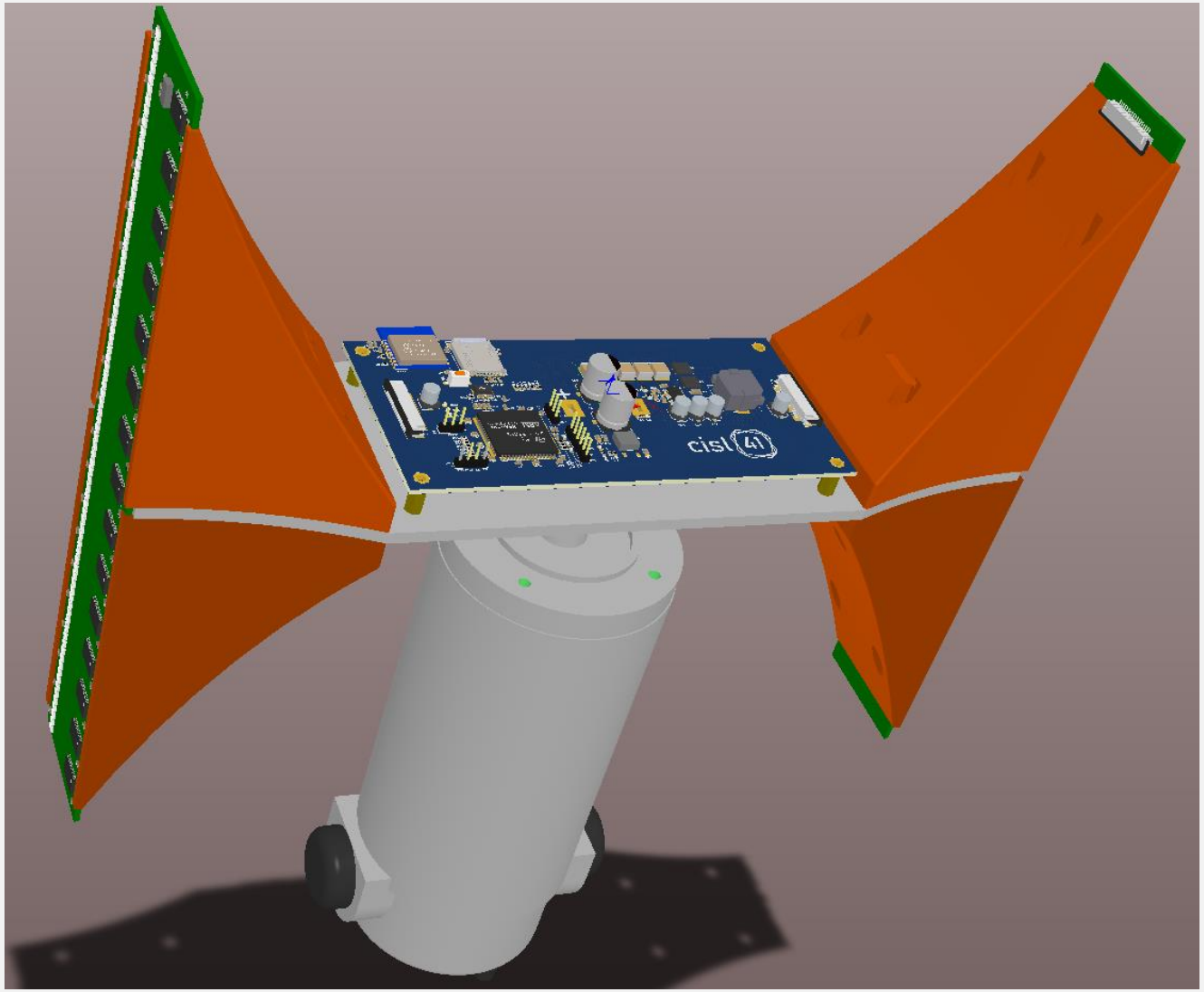
© https://www.freepik.com/vectorpouch

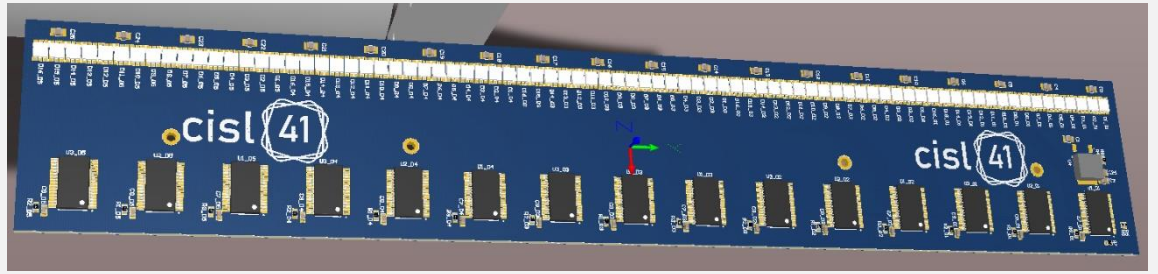
Exemple PCB custom

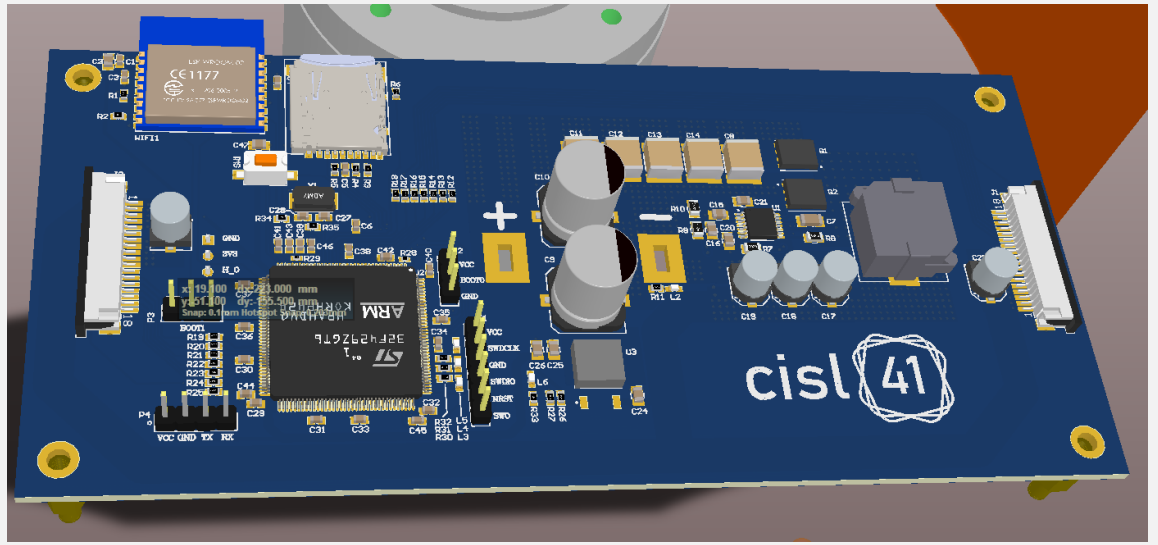




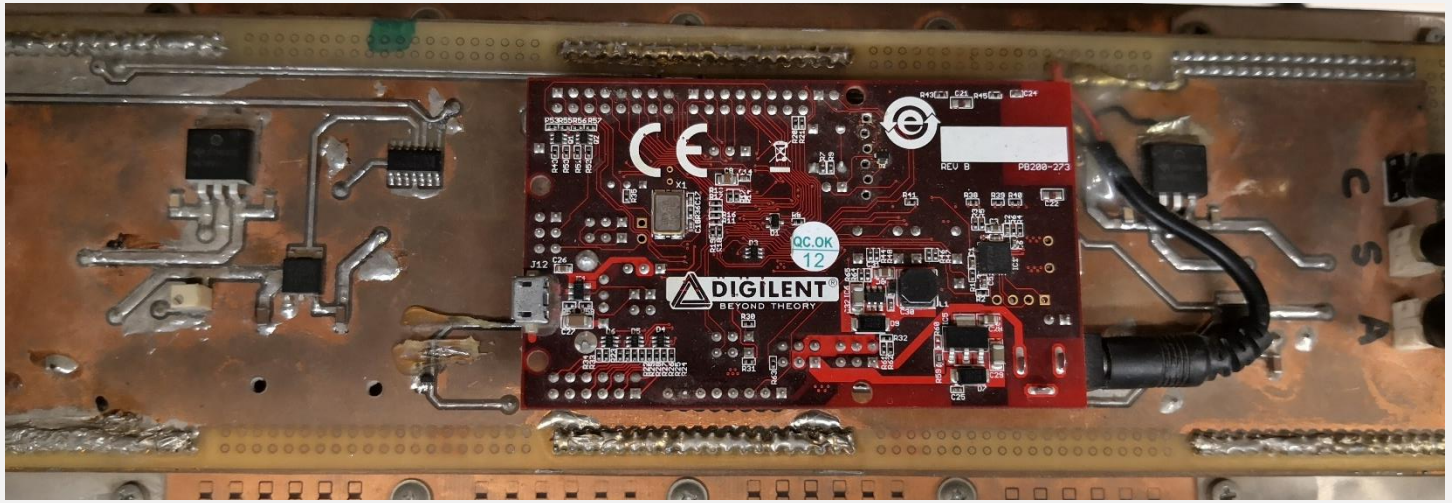





cisl 41
cisl 41

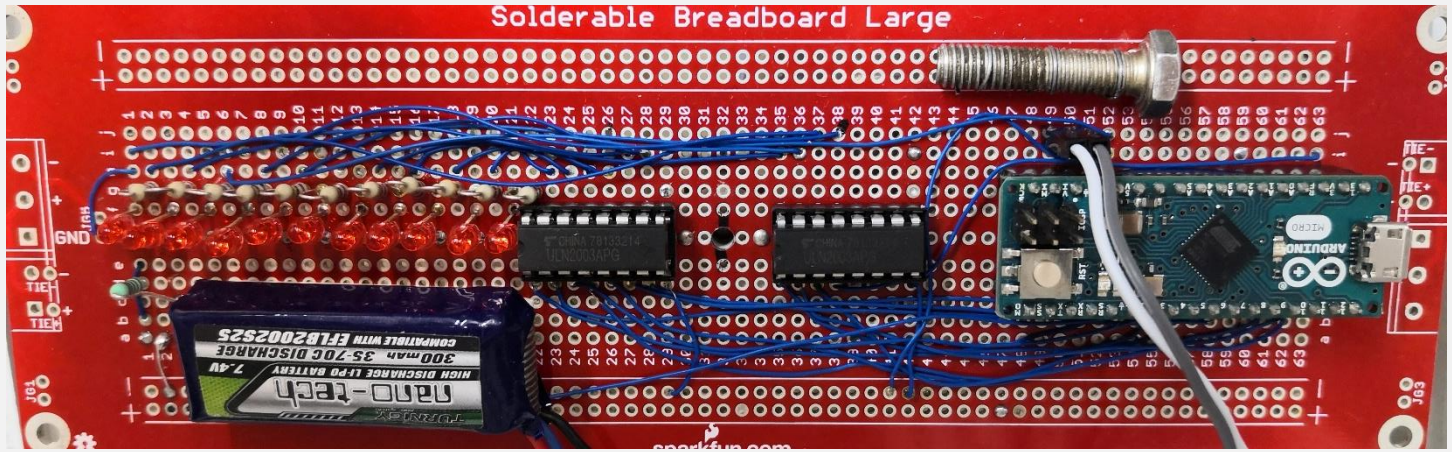

cisl 41

Mixt design



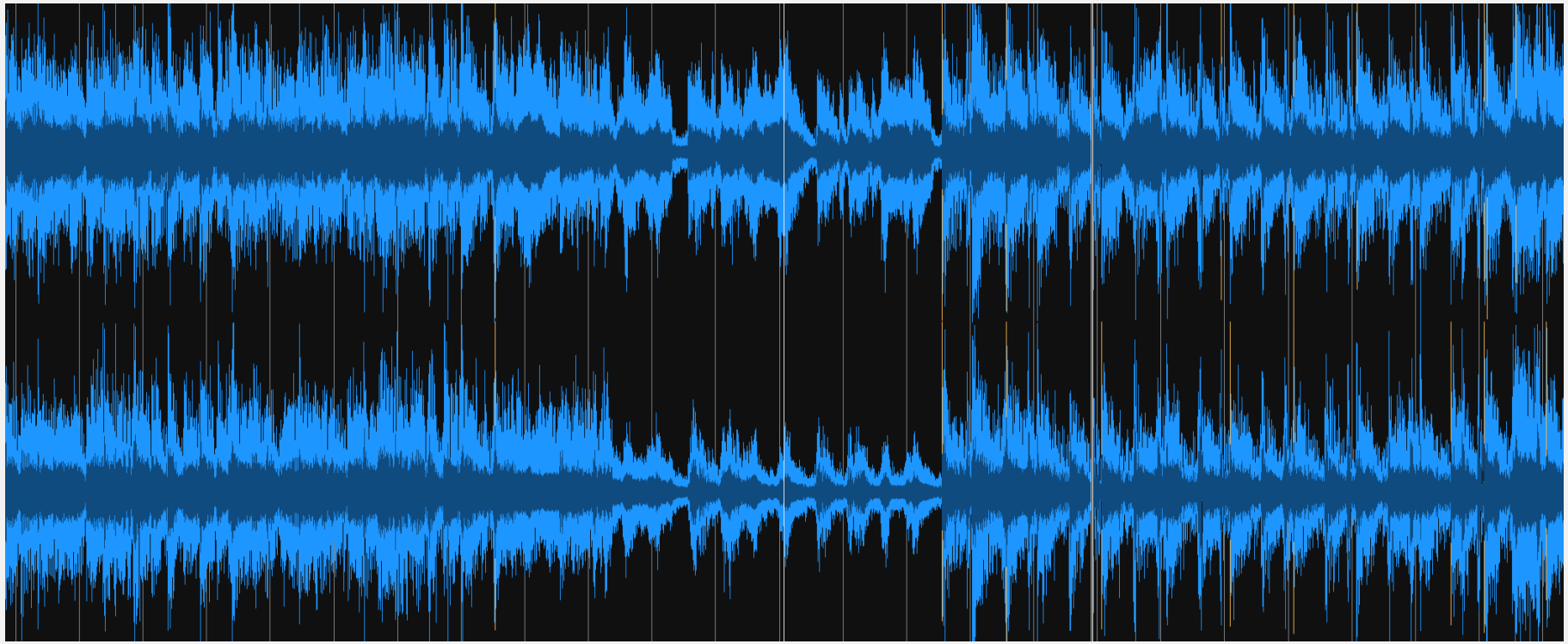


Breadboard



Câteva definiții

- semnal - fenomen (electric) / mărime (electrică) ce reprezintă informație (analogic/ digital)



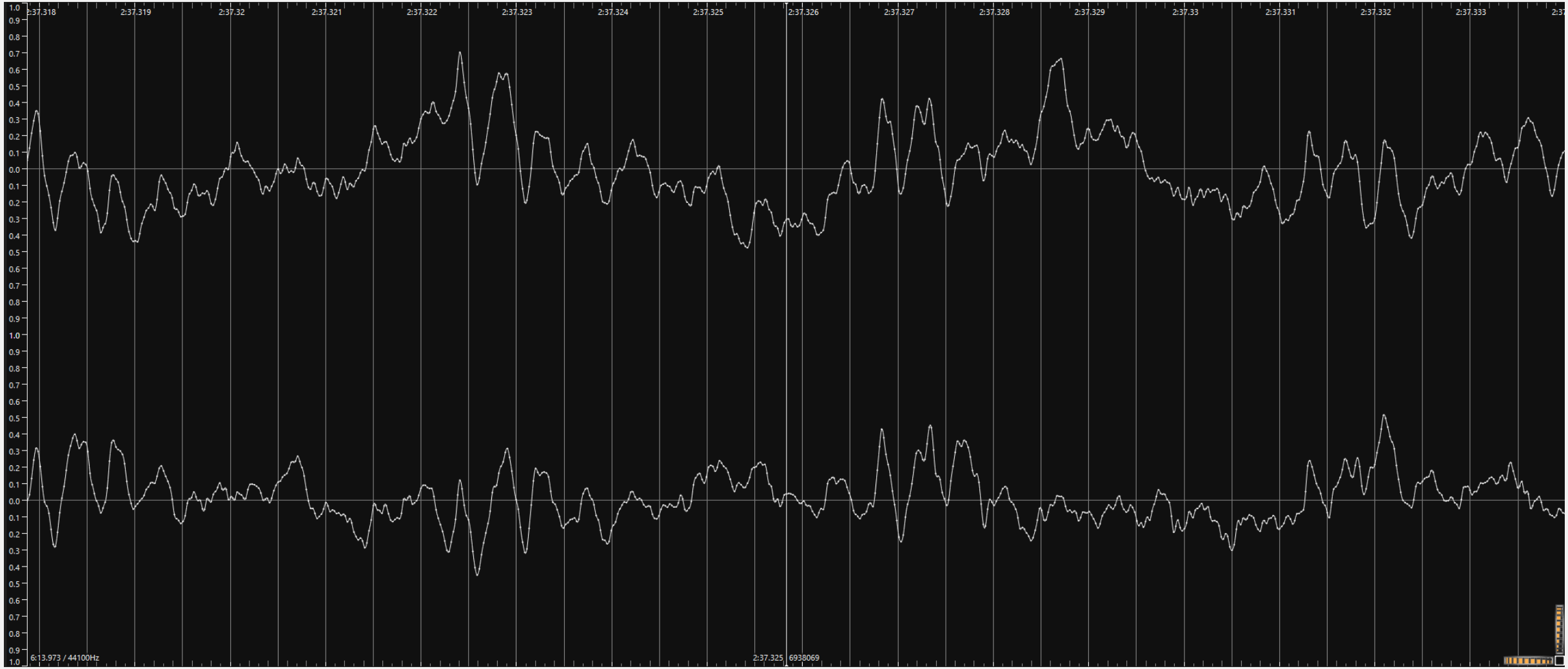
De ce “Analogică”

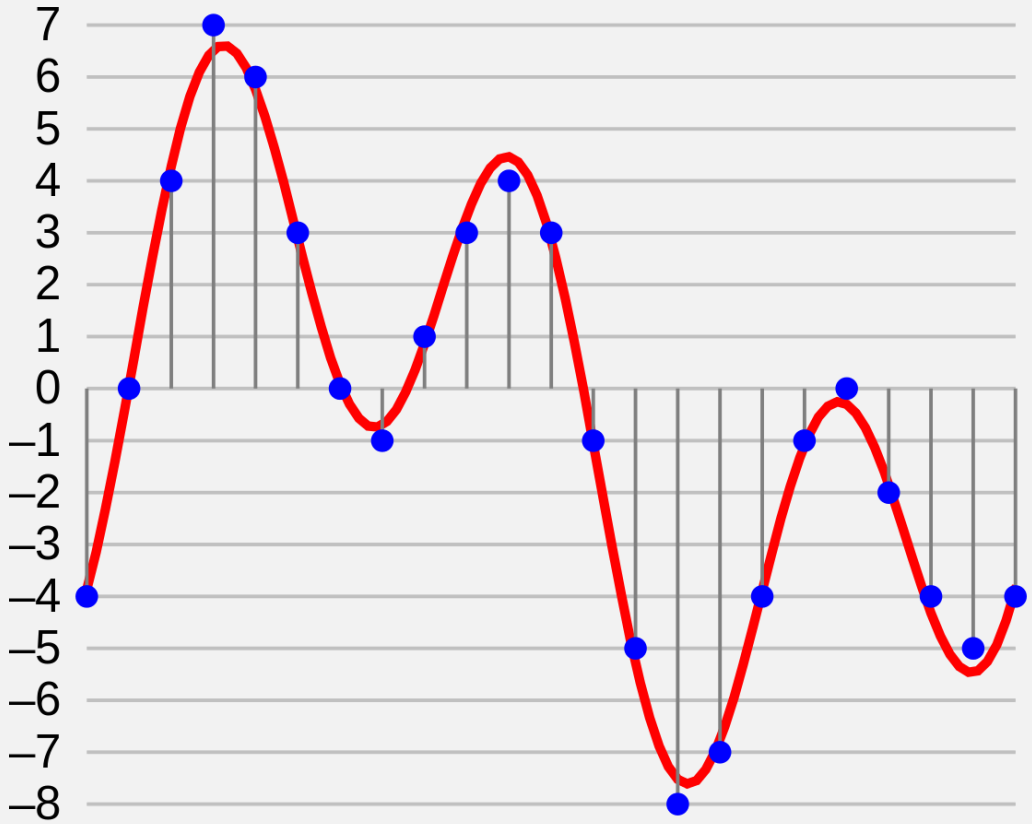


ανάλογος

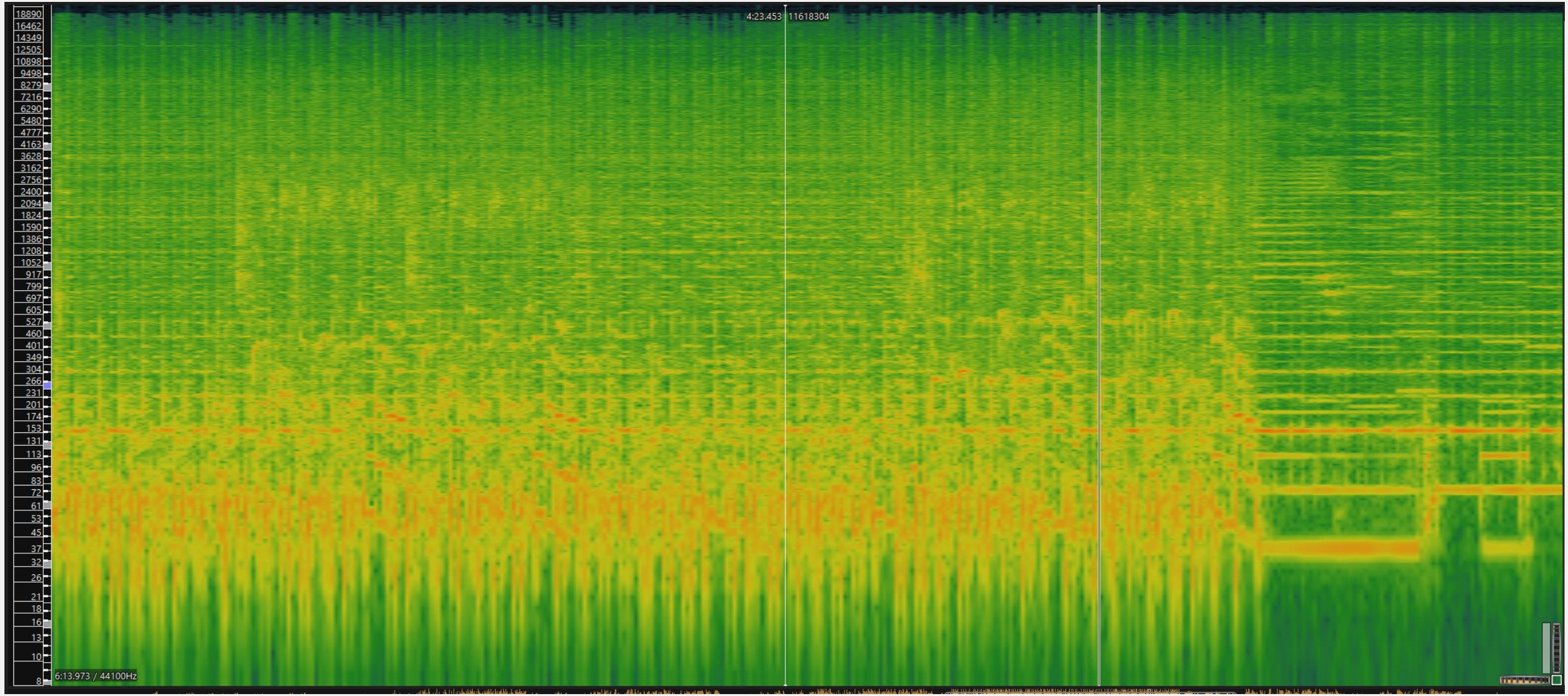
analogos (proporțional)

De ce “Analogică”





© https://en.wikipedia.org/wiki/Digital_audio#/media/File:4-bit-linear-PCM.svg


4:23.453
11618304
18890
16462
14349
12505
10898
9498
8279
7216
6290
5480
4777
4163
3628
3162
2756
2400
2094
1824
1590
1386
1208
1052
917
799
697
605
527
460
401
349
304
266
231
201
174
153
131
113
96
83
72
61
53
45
37
32
26
21
18
16
13
10
8
6:13.973 / 44100Hz

Exemple semnale analogice

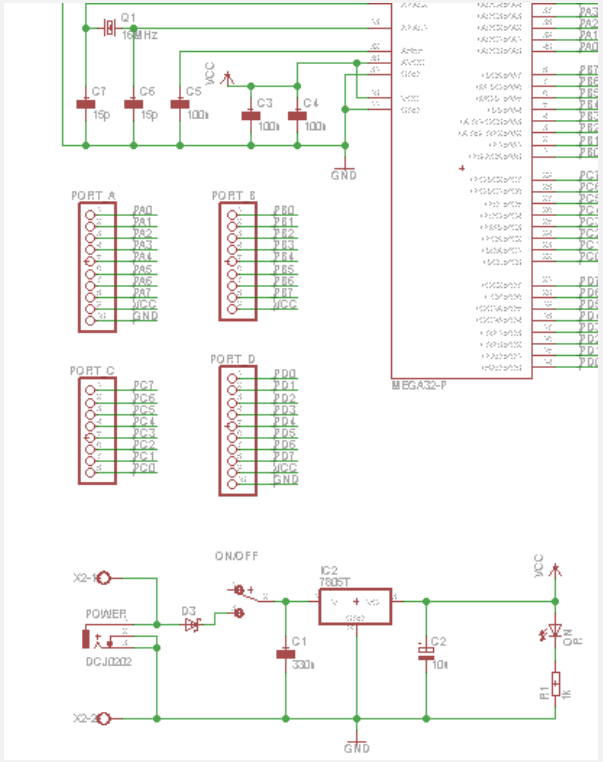
- Sunet
- Temperatura
- Presiune
- Acceleratie
- Forță
- Viteză
- Intensitate luminoasă

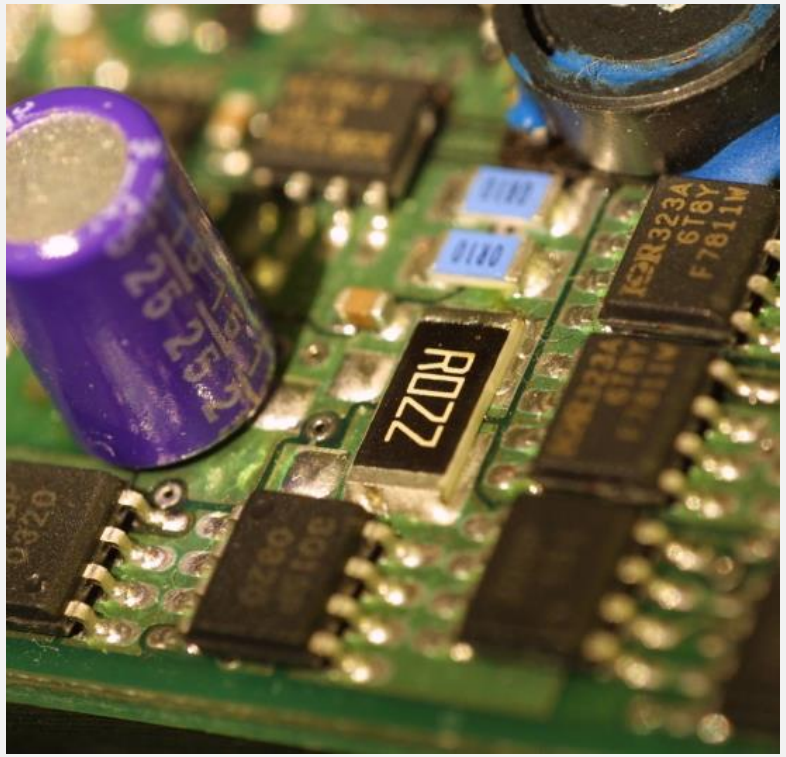
Reprezentare info digital vs analogic

- obs: input/ output – in general A
- Digital:
  - precizie redusă – atât în amplitudine cât și în timp
  - (o data cu compresia) – pierderi mai mari
- Analogic
  - în mod practic, plaja de valori limitată
  - zgomot
  - variații funcție de condiții & starea circuitului

Obiective curs







Exemple companii - angajator (în RO)



Cursul următor..

Curs 02

- Recapitulare ☺